

The Science of Art Issue — Artometrics

The Science of Art Issue

Canon, museums, heritage, and the historical bridge from artistic practice to scientific method.

Who Gets Remembered in the Global Pantheon of Fame?

Collective memory is not a neutral archive. It is a ranked attention system: some lives travel across languages and centuries, while others remain local even when their achievements were large.

Pantheon-style biography indexes — on the order of **85K+** lives in public-facing summaries, with a collection logic that requires presence in at least **15** Wikipedia language editions — make that ranking measurable. The 2016 *Scientific Data* paper for Pantheon 1.0 is the scholarly anchor. The charts below treat three main memory engines — power, art, and science — across **10** memory-capital cities.

If humanity remembers some kinds of people more than others, that pattern is itself a cultural object. The question is which occupations and cities compound fame, and which leave gaps.

Fast facts

The numbers that set the scale for this report:

85K+Biographies indexed by Pantheon in public-facing summaries

15Minimum Wikipedia language editions used in Pantheon collection logic

2016Scientific Data paper for Pantheon 1.0

3Main memory engines: power, art, science

10Memory-capital cities compared

Data and method

Pantheon aggregates biographical notability using multilingual Wikipedia presence and related fame metrics. The dataset is powerful because cross-language edition counts act as a rough passport stamp for global memory — a person who exists in many Wikipedias has traveled farther through culture than a person who exists in one.

The charts here use editorial indices to make domain, era, city, and occupation comparisons legible. They are interpretive maps grounded in Pantheon's public logic, not a claim that Wikipedia is the only archive that matters.

Memory domains

Power, art, and science travel farther across collective memory

Politics, arts, science, sports, business, religion, exploration, and activism do not travel equally through collective memory. Power, art, and science repeatedly show up as the main engines that cross languages and centuries.

Sports and business can be enormously famous in the present and still underperform on long multilingual durability. The domain chart separates contemporary visibility from archival carry.

Era shift

The remembered person changes from ruler to cultural or scientific producer

Across eras — ancient, medieval, early modern, 1800s, 1900s, 2000s — the remembered person changes. Rulers dominate earlier memory shares; cultural and scientific producers rise as print, universities, and later mass media create new fame factories.

The shift is not a morality play about progress. It is a change in which institutions mint durable names: courts and churches first, then academies and culture industries.

Memory capitals

Cities become archives when institutions compound across centuries

Paris, London, Rome, New York, Vienna, Berlin, Athens, and Moscow appear among the memory-capital comparisons because institutions compound there: universities, courts, publishers, studios, academies. Cities become archives when those institutions last long enough to keep minting and storing notable lives.

A memory capital is not identical to a current GDP capital. Some cities punch above their present economic weight because their historical institutions still feed the multilingual record.

Two clocks of fame

Some occupations have durable translation while others spike through present attention

Some occupations show durable translation across languages; others spike through present attention without the same archival carry. Literature, film, music, science, sports, and politics sit on different clocks — one measured in centuries of citation, one measured in contemporary noise.

The two-clock chart is the practical tool for editors and educators: it warns against treating a viral profession as automatically canonical, and against treating a canonical profession as automatically popular.

Archive gap

The archive remembers through historical inequality

The archive remembers through historical inequality. Domains and regions with denser institutions, denser literacy, and denser Wikipedia coverage look more important because they are more recorded.

That gap is not a reason to discard Pantheon. It is the reason to read it as a map of remembered inequality — a cultural object that reveals what global memory systems have been able to store.

What to take away

Collective memory, measured through Pantheon-style multilingual biography indexes, concentrates on power, art, and science, compounds in a short list of memory-capital cities, and runs on two clocks of fame.

The citable claim is modest: what humanity remembers at global scale is structured — by occupation, era, and city — and that structure is itself evidence about cultural power.

Sources

Where UK Museums Sit on the Deprivation Map

Britain's museum map is also a map of place. This file places 4,191 museum records against an area deprivation index whose median is 5.00 and whose observed high is 10.0. Blenheim Palace appears among the fact-box notables; the South West of England shows up in the geographic gap analysis.

The question is not which museum is “best.” It is how cultural institutions sit on a deprivation scale — whether collections cluster in less-deprived areas, and how far named sites and admin areas sit above or below the median index of 5.00.

Fast facts

The numbers that set the scale for this report:

4,191Records in the working dataset

5.00Median Area Deprivation index

10.0Highest observed Area Deprivation index

Blenheim PalaceTop Name of museum by Area Deprivation index

/England/South West (EnglishMost common Admin area

Data and method

The source is the TidyTuesday release from 2022-11-22 (R for Data Science community). The working file contains 4,191 rows and 35 columns after merging available tables in the week folder. Area deprivation index is the primary metric; museum name and admin area are the main categorical axes; DOMUS identifier appears in the scatter.

Medians are used for robustness. Index-style fields are excluded from metric selection. Deprivation scores describe the surrounding area context in the source, not an editorial ranking of museum quality.

Named museums pin the top of the deprivation scale

Area Deprivation index by Name of museum

In the name breakdown, The Archaeology Centre leads at 10.0; Smithy Heritage Centre anchors the low end of that plotted group at 10.0 as well — a reminder that the upper slice of this particular cut can compress when many sites share the same index band.

Grouping by museum name makes individual institutions visible on the deprivation axis. It is a catalog of places, not a visitor-numbers league table.

The leaders chart sits at the index ceiling

Top Name of museum

The Archaeology Centre leads at 10.0, and 10.0 also marks the median among the top dozen in this cut. When the top of the ladder is flat, the story is saturation at the high end of the deprivation index rather than a single runaway institution.

Blenheim Palace remains a fact-box landmark for the file; the charted leaders emphasize which names hit the observed maximum of 10.0 in this aggregation.

Admin areas spread museums across different deprivation bands

Area Deprivation index by Admin area

Category boxes by admin area show whether deprivation consensus is shared or contested across local jurisdictions. Museums do not all sit in the same band; geography fragments the index.

That spread is the point of an atlas-style reading: cultural infrastructure is unevenly distributed across deprivation contexts, and the boxes make the unevenness visible.

Wiltshire clears the median; Cornwall trails in the South West cut

Area Deprivation index vs median by Admin area

/England/South West (English Region)/Wiltshire (English UA) sits 2.00 above the median; /England/South West (English Region)/Cornwall (English UA) trails by 1.00. Even inside one English region, admin areas do not share the same offset from the file center.

Signed gaps turn geography into a comparable scale. They do not explain why a museum is where it is; they locate the area context relative to the median deprivation index of 5.00.

Deprivation and DOMUS IDs form a joint map

Area Deprivation index vs DOMUS identifier

Plotting area deprivation index against DOMUS identifier shows clusters that a single average erases. Identifier ranges and deprivation bands co-locate in ways that matter for joining and auditing the catalog.

The scatter is a data-structure check as much as a social map: it shows how the numeric deprivation field sits beside the museum identity field in the 4,191-row extract.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live heritage registries. Missing values, naming variants, and week-of-export coverage limits apply. Merged tables may fan out or duplicate rows when join keys are imperfect.

Area deprivation index describes place context in the source file. Findings are structural signals about UK museums in this extract — not a verdict on institutional excellence or funding.

What to take away

UK museums in this file sit around a median deprivation index of 5.00, with an observed high of 10.0 and clear admin-area gaps even inside the South West.

Name leaders, area boxes, and the deprivation–DOMUS scatter together show how cultural sites map onto inequality of place — a geography story first, a prestige story only by implication.

Sources

Data Science Learning Community. (2022). *TidyTuesday: UK Museums*.
<https://raw.githubusercontent.com/rfordatascience/tidyuesday/main/data/2022/2022-11-22/museums.csv>

Editor's note

Artometrics data report from the TidyTuesday research pipeline. Charts and aggregates are reproducible from the embedded exhibits and public source files.

[View TidyTuesday source on GitHub](#)

How UNESCO Sites Grew Across Scandinavia, 2004–2022

UNESCO World Heritage counts turn national prestige into a time series. This Scandinavian extract holds 6 records spanning 2004–2022, with a median value of 9.00 and a high of 15.0. Sweden leads the fact-box country ranking.

Six rows are enough for a clean comparative story when the table is country-by-year counts: who led, who grew fastest, and how the median moved from 5.00 to 10.0 across the window. The calibration point is 9.00 — the center of this small but sharp heritage ledger.

Fast facts

The numbers that set the scale for this report:

6Records in the working dataset
9.00Median Value
15.0Highest observed Value
SwedenTop Country by Value
2004–2022Year span covered in the file

Data and method

The source is the TidyTuesday release from 2024-02-06 (R for Data Science community). The working file contains 6 rows and 3 columns after merging available tables in the week folder. Value is the site-count metric; country is the entity axis; year compares 2004 with 2022.

With only six rows, every chart is close to the raw table. Medians still summarize the center; growth percentages normalize different starting points. Index-style fields are excluded from metric selection.

Median heritage counts doubled at the center

Median Value Over Time

Median value rises from 5.00 in the opening period to 10.0 at the close. In this extract, the center of the Scandinavian heritage-count distribution moved upward across 2004–2022.

A rising median in a six-row file is still a real trend — it is simply a trend you can nearly audit by eye against the country panels.

Sweden leads the country ladder

Sweden leads at 14.0 — 7.00 marks the median among the top dozen

Sweden leads at 14.0, while 7.00 marks the median among the top dozen in this cut — here, effectively the small peer set of Scandinavian countries in the file. Sweden is both the fact-box top country and the charted leader.

Leadership on level is not the same as leadership on growth. The later charts separate stock from flow.

Leaders diverge when tracked over time

Top Country Over Time

Leader trends show that leading names do not move in lockstep — some fade as others surge. Tracking value over time separates sustained dominance from one-off jumps.

In a short panel, trajectory is the whole game: who added sites between the two observed years, and who merely started ahead.

2004 versus 2022 makes the gains visible side by side

Value by year and Country

Grouped bars expose who gained between 2004 and 2022 — not just the latest leaderboard. Small panels reward side-by-side reading; totals alone hide per-country momentum.

The year compare is the clearest view of absolute change before converting those changes into percentages.

Denmark posted the largest percentage gain

Value growth by Country

Denmark posted the largest gain (150%) from 2004 to 2022. Percent-change bars normalize different starting points — essential when baselines differ across Sweden, Norway, and Denmark.

Sweden can lead on level while Denmark leads on growth. Both facts are in the file; they answer different heritage-expansion questions.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not the live UNESCO World Heritage List API. This extract is a six-row Scandinavian panel for 2004 and 2022, not a global inventory of every site.

Findings describe this world-heritage extract — structural signals about country counts and growth — not a qualitative ranking of sites' cultural importance.

What to take away

Scandinavian UNESCO counts in this file rise at the median from 5.00 to 10.0, with Sweden leading on level at 14.0 and Denmark leading on growth at 150% from 2004 to 2022.

Level and growth are different trophies. The year-compare and growth charts keep them separate so prestige stock is not mistaken for expansion speed.

Sources

Data Science Learning Community. (2024). *TidyTuesday: World Heritage Sites*. <https://raw.githubusercontent.com/rfordatascience/tidyuesday/main/data/2024/2024-02-06/heritage.csv>

Editor's note

Artometrics data report from the TidyTuesday research pipeline. Charts and aggregates are reproducible from the embedded exhibits and public source files.

[View TidyTuesday source on GitHub](#)

Which Subjects Dominate the Project Gutenberg Shelf?

Project Gutenberg is the internet's oldest large-scale experiment in free literature. Its subject catalog — Library of Congress headings and related labels — is a map of which public-domain books volunteers could digitize and classify at scale.

A TidyTuesday working extract of **100,000** records shows **lcsh** as the most common subject type, with tens of thousands of rows in that bucket. Subject codes such as **PS** recur thousands of times. The catalog is not every book ever written; it is the reusable shelf that copyright expiry, volunteer labor, and classification systems made easy to share.

Reading Gutenberg as a popularity contest misses the point. Reading it as a concentration map of reusable canon is where the data help.

Fast facts

The numbers that set the scale for this report:

100,000Records in the working dataset
lcshMost common Subject type

Data and method

The source is the TidyTuesday Project Gutenberg release from the R for Data Science community. The working file contains 100,000 rows after assembly — subject types, subject labels, and related catalog metadata.

Because many fields are categorical, the analysis leans on counts and repetition rather than a single quality score. Charts export as Plotly JSON with PNG fallbacks. Subject headings are librarian infrastructure, not reader reviews.

Public-domain books cluster by subject type

Public-domain books cluster by subject type

lcsh dominates with on the order of **69,027** records in the highlighted cut, far ahead of thinner subject-type buckets. The main classification family carries the story; this field does not split into many equal long-tail types.

That concentration means most navigational claims about what Gutenberg contains are really claims about how LCSH-style headings organize the digitized shelf.

A small set of subjects anchors the catalog

A small set of subjects anchors the catalog

PS appears about **4,684** times — among the most recurring subject names in the file. The top dozen account for a visible share of all 100,000 rows even though most subject entities appear only once.

American literature codes, fiction labels, and related headings form a reusable core. They are the catalog's gravitational center for teachers, scrapers, and adaptation hunters alike.

Subject families show the catalog center of gravity

Subject families show the catalog's center of gravity

lcsh is again the largest bucket on the category chart. Subject families show where editorial attention should focus first if the goal is to understand the shelf's center of gravity rather than its exotic edges.

The edges still matter for discovery. They do not define the statistical middle of a 100,000-row extract.

Repeated subjects reveal the reusable canon

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Most subject entities appear only once; a small head revisits repeatedly. That power-law shape is typical of catalog tables: a reusable canon of headings, and a long inventory of singleton classifications.

Repeated subjects are the ones most likely to support classroom packs, themed collections, and machine-learning corpora. Frequency is a reuse forecast as much as a shelf description.

Subject labels become the map of the shelf

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PS and related labels become the map of the shelf when numeric scores are sparse. Frequency leaders reveal franchise depth in literature the way studio logos reveal franchise depth in film.

The practical claim for cultural analytics is simple: if you can only afford to study a slice of Gutenberg, the repeated subject head is where coverage of the reusable public-domain canon begins.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and sampling limits apply. Subject headings are not sales, downloads, or reading-time proof.

Findings describe structural signals about Project Gutenberg subject metadata — not a complete history of world literature, and not a ranking of artistic merit.

What to take away

Gutenberg's subject catalog is concentrated: lcsh dominates subject type, a short head of codes such as PS recurs thousands of times, and most labels appear once.

The citable lesson is about reusable canon. Public-domain literature becomes infrastructure when classification and digitization make the same subjects easy to find again and again.

Sources

Mapping the Public-Domain Canon Through Project Gutenberg

Project Gutenberg is not only a library. It is a map of what digitized public-domain culture looks like when volunteer labor, copyright timers, and language markets decide what is easy to share.

Public-facing summaries place the collection on the order of **75K+** ebooks, with catalog feeds updating on **weekly** CSV and **daily** RDF cadences in Gutenberg tooling docs. The first chart compares **8** languages; later panels use **10** authors as anchors for availability and memory.

English gravity, nineteenth-century density, and adaptation-ready subjects are the three forces that turn a free shelf into a cultural operating system for schools, remixers, and machines.

Fast facts

The numbers that set the scale for this report:

75K+ Approximate scale of Project Gutenberg ebooks in public-facing summaries

Weekly CSV catalog update cadence noted in Gutenberg tooling docs

Daily RDF catalog update cadence noted by Gutenberg

8 Languages compared in the first chart

10 Authors used as anchors

Data and method

Project Gutenberg provides machine-readable metadata in RDF/XML, MARC, and CSV formats. Official pages recommend using the metadata feeds rather than scraping HTML. A scaled production pass should ingest the weekly CSV or RDF catalog, normalize subjects, languages, and authors, and join to adaptation, syllabus, and Wikidata signals.

The charts here use editorial indices to define that comparative structure before a full catalog join. They are meant to make language gravity, era mix, author memory, and remix power legible — not to replace the official feeds.

Language gravity

English dominates the accessible public-domain shelf

English dominates the accessible public-domain shelf in the comparison of eight languages — on the order of 72 on the index scale used in the chart, with French and German far behind and Finnish, Dutch, Portuguese, Italian, and Spanish in a thinner band.

Language gravity is not a judgment about literary quality. It is a statement about which texts are easiest to search, remix, teach, and feed into downstream culture when copyright and digitization align.

The era machine

The 19th century becomes the public-domain literary core

Copyright law turns time into a cultural filter. Fiction and literature indices rise sharply through the nineteenth century; nonfiction and reference follow a related but not identical path. Pre-1800 and post-1950 bins look thinner on the reusable shelf.

That makes Gutenberg a meso dataset: not every book ever written, but the books most ready to be reactivated because they are old enough, popular enough, and digitized enough.

Author memory

Digital availability and cultural memory reinforce each other

Authors become infrastructure when their works are everywhere: classrooms, editions, audiobooks, adaptations, quote databases, and training corpora. Digital availability and cultural memory reinforce each other rather than acting as independent clocks.

That feedback loop is why public domain matters to AI culture too. What is easy to load is what models and people alike can easily learn from.

Adaptation power

Adventure, childhood, and gothic subjects adapt especially well

Adventure, manners, gothic, science, war, travel, religion, and children's subjects do not adapt equally. Adventure, childhood, and gothic titles move especially cleanly across film, brand memory, and genre grammar.

The public-domain market is therefore not dead literature. It is reusable cultural material with uneven conversion rates into new media.

Canon and remix

Some books become curriculum while others become remix engines

Some books become curriculum while others become remix engines. *Pride and Prejudice* is classroom canon and adaptation fuel. Sherlock Holmes, Dracula, and Frankenstein behave more like cultural APIs: infinitely forkable characters and premises.

The strongest public-domain works are not only read. They are reused — assigned, quoted, adapted, and recombined until availability becomes a kind of fame.

What to take away

The public-domain canon is not neutral. It is shaped by language, copyright age, volunteer digitization, educational reuse, and adaptation economics.

Gutenberg is a bridge dataset: it connects literature, AI training culture, education, film adaptation, and historical memory through the same shelf of reusable texts.

Sources

How Concentrated Is the World's Linguistic Diversity?

Linguistic diversity is often imagined as a smooth global gradient. Glottolog-style language inventories show something lumpier: documentation clusters by macroarea, and a long tail of languages appear once while a short head of names and identifiers repeats.

The TidyTuesday working file holds **8,612** records. **Africa** is the most common macroarea label, with **2,363** records in that bucket alone. The question is not only how many languages exist — it is where documentation effort, and therefore visibility, concentrates.

A catalog of languages is also a catalog of scholarly attention. Places with denser fieldwork traditions look richer in the table even when the underlying human diversity story is more even — or more endangered — than the row counts imply.

Research question

What does a Glottolog-derived catalog reveal about the geography of linguistic documentation, not just the geography of language itself? This report asks where the 8,612-record extract concentrates by macroarea, where names and identifiers repeat, and how much of the visible language map is shaped by catalog infrastructure.

The question is observational because row counts are not speaker counts, vitality scores, or political claims. The aim is to distinguish language diversity as lived by communities from language visibility as preserved in databases, grammars, ISO-style identifiers, and scholarly citation systems.

Fast facts

The numbers that set the scale for this report:

8,612Records in the working dataset
AfricaMost common Macroarea

Data and method

The source is the TidyTuesday release from 2025-12-23 (R for Data Science community). The working file contains 8,612 rows and 9 columns after cleaning — language names, identifiers, macroarea labels, and related Glottolog metadata.

Because many fields are categorical rather than scored, the analysis leans on counts, concentration, and repetition rather than medians of a single numeric quality metric. Charts export as Plotly JSON with PNG fallbacks.

A geographically uneven landscape

Language documentation is geographically uneven

Africa dominates with **2,363** records in the primary macroarea cut. Other macroareas — Papunesia, Eurasia, the Americas, Australia — follow at lower counts. The main bucket carries much of the story; this field does not behave like a gentle long-tail split across dozens of equal regions.

Uneven documentation is not identical to uneven linguistic diversity, but it shapes what comparative science can see. Extinction risk, revitalization, and typology research all inherit the map of what has been recorded.

Africa's lead is plausible in linguistic terms: the continent contains large families such as Niger-Congo, Afro-Asiatic, Nilo-Saharan groupings as traditionally defined, and Khoisan-associated lineages, alongside intense contact zones and multilingual states. But the chart does not validate any particular family taxonomy; it records how the Glottolog-style file assigns macroarea metadata to rows.

Papunesia and Australia are especially important comparison cases because they contain many small language communities with high typological diversity and severe endangerment pressure. A lower row count than Africa does not mean lower scientific importance. It means the record is distributed unevenly across regions, fieldwork histories, and naming conventions.

Who sits at the top of name repetition

Repeated language names show documentation density

Repeated language names show documentation density inside the extract. **Fasu** appears among the recurring names in the file; the top dozen account for a visible share of all 8,612 rows even though most languages appear only once.

Name repetition can reflect dialect documentation, alternate glottonyms, or multiple database entries tied to related varieties. It is a signal to investigate, not an automatic claim that one language is more important.

Fasu is a Papuan language name associated with Papua New Guinea, and its recurrence is a reminder that catalog entries are not always one-to-one with what a lay reader calls a language. Varieties, dialect clusters, alternate names, bibliographic records, and genealogical nodes can all push the same surface name into repeated view.

This is where Glottolog's infrastructure differs from a census. A national census might ask who speaks a language at home; Glottolog asks how a language or lineage is classified and documented in the linguistic literature. The same community can be highly visible in one system and nearly invisible in another.

Regional concentration

Regional concentration shapes the language map

Africa is again the largest bucket with 2,363 records on the regional concentration chart. Category concentration shows where editorial and scientific attention should look first if the goal is to understand the file's center of gravity.

Mixed macroarea labels in the data — combinations such as Eurasia with Papunesia — remind readers that languages and contact zones do not always respect tidy continental borders.

The macroarea scheme is a practical database compromise. Eurasia contains Indo-European, Uralic, Turkic, Sino-Tibetan, Japonic, Koreanic, Dravidian, and many other lineages; the Americas contain hundreds of Indigenous languages with colonial-era documentation gaps; Papunesia compresses a spectacularly dense set of Papuan and Austronesian histories into one label. The labels help chart structure, but they are not theory in themselves.

Regional concentration also encodes academic history. Missionary grammars, colonial surveys, national language academies, SIL International work, university field programs, and contemporary community-led documentation all leave traces in catalogs. The chart is therefore a map of evidence production as much as a map of human speech.

Most languages appear once

Most languages appear once while a small head repeats

Most name entities appear only once; a small head revisits repeatedly. That power-law shape is typical of catalog-style tables: a few repeatedly documented varieties, and a long inventory of singleton entries.

For endangered-language work, the singleton majority is the point. Visibility in a database is already a scarce resource; languages that appear once may still be under-described relative to their social importance.

A singleton entry can represent a language with thousands of speakers, a severely endangered variety with only elders remaining, or a poorly documented name whose classification remains uncertain. Without speaker counts, vitality levels, and bibliographic depth, the frequency chart should be read as documentation structure, not community scale.

UNESCO's language vitality framework and Ethnologue-style speaker estimates answer different questions than Glottolog identifiers. The ArtoMetrics chart is valuable precisely because it shows the catalog layer that those other systems need to join against before scholars can ask about endangerment, education policy, or intergenerational transmission.

Identifiers as metadata

Identifier fields are metadata rather than a reader-facing thesis

Glottolog-style IDs such as **fasu1242** are the most repeated identifiers in the extract. Secondary dimensions add context when the primary table has no numeric score column — they are join keys for typology databases, not a popularity contest.

Reading IDs as if they were rankings misunderstands the infrastructure. The useful claim is that documentation systems create machine-readable handles that make some languages easier to cite, map, and study than others.

Identifier fields are the quiet machinery of comparative linguistics. Glottocodes such as **fasu1242**, ISO 639-3 codes, and family IDs let researchers connect grammars, lexicons, sound inventories, phylogenies, and geographic coordinates without relying on unstable spelling variants. The identifier is not the language; it is the database handle that lets the language remain findable.

That infrastructure has real consequences. A language with a stable identifier can be linked into typological databases such as WALS, Glottolog, PHOIBLE, or Grambank more easily than a language whose names are fragmented across archives. Machine readability becomes a form of scholarly visibility.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live APIs. Missing values, spelling variants, and release coverage limits apply. Row counts are not speaker counts, and macroarea labels are not political maps.

The file cannot by itself measure vitality, endangerment, or intergenerational transmission. It measures documentation structure — a necessary but incomplete layer of the global language story.

What to take away

World linguistic diversity, as seen through this Glottolog-derived extract, is geographically lumpy: Africa leads the macroarea counts at 2,363 of 8,612 records, while most language names appear once.

The citable lesson is about visibility. What scholars can count depends on where fieldwork and cataloging have been dense — and that density is itself a cultural and scientific fact.

Sources

Data Science Learning Community. (2025). *TidyTuesday: Glottolog / language data*. <https://github.com/rfordata-science/tidy-tuesday/tree/main/data/2025/2025-12-23>

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Which U.S. PhD Fields Expanded Fastest?

U.S. doctorate production is a labor-market story told in field counts. This file holds 3,370 records spanning 2008–2017, with a median of 85.0 PhDs and a high of 5,302. Social sciences lead the fact-box field ranking; life sciences appear among the notable broad fields.

The question is which disciplines grew — and which sit above or below the median production level — when doctorate output is measured year by year and field by field. Medians keep a few giant programs from dictating every comparison.

Fast facts

The numbers that set the scale for this report:

3,370Records in the working dataset
85.0Median N phds
5,302Highest observed N phds
Social sciencesTop Field by N phds
2008–2017Year span covered in the file
Life sciencesMost common Broad field

Data and method

The source is the TidyTuesday release from 2019-02-19 (R for Data Science community). The working file contains 3,370 rows and 6 columns after merging available tables in the week folder. N phds is the primary metric; broad field is the main categorical axis.

Medians are used because field sizes skew. Index-style fields are excluded from metric selection. Charts track trend, leaders, distribution, gaps to the median, and concentration across fields.

Median doctorate counts edged upward across the decade

Median N phds Over Time

Median n phds rises from 80.0 in the opening period to 87.0 at the close — a modest climb around the file median of 85.0. The typical field-year in this extract produced slightly more doctorates by the end of the window than at the start.

A rising median is not the same as uniform growth. Leader fields can surge while smaller programs move little; the trend chart reports the center, not every discipline's path.

Social sciences lead the field ladder

Social sciences leads at 4,944 — 915 marks the median among the top dozen

Social sciences leads at 4,944 PhDs in the leaders cut, while 915 marks the median among the top dozen. The distance between first place and that top-dozen median shows how quickly the ladder drops even inside the upper tier.

Fact-box highlights place social sciences at the top of the field ranking and note life sciences among the broad-field landmarks. The chart makes the numeric gap concrete.

Broad fields do not share one production band

N phds by Broad field

Category boxes by broad field show whether doctorate counts are shared or contested across disciplinary families. Psychology and social sciences, engineering, life sciences, and others occupy different parts of the distribution.

The boxes are the right tool when fields differ in scale. A single national median of 85.0 hides those family-level spreads.

Psychology and social sciences clear the median; engineering trails

N phds vs median by Broad field

Psychology and social sciences sits 30.5 above the median; engineering trails by 21.5. Those signed gaps convert broad-field boxes into distance from the file center.

Trailing the median is not a quality judgment. It is a statement about relative volume in this extract's n phds metric.

Five fields hold most of the aggregate doctorates

Cumulative N phds

The top five field entries account for 63% of the aggregate n phds. Concentration is high: a small set of fields drives most of the summed doctorate production in the file.

Steep Pareto curves mean capacity and attention cluster. The long tail of smaller fields still matters for the ecosystem, but it does not dominate the total.

What this file cannot tell you

Community-cleaned TidyTuesday snapshots are not live NSF Survey of Earned Doctorates APIs. Missing values, field-name variants, and 2008–2017 coverage limits apply. Merged tables may fan out or duplicate rows when join keys are imperfect.

Findings describe this U.S. PhDs extract — structural signals about doctorate counts by field — not a full labor-market forecast or ranking of program quality.

What to take away

Doctorate production in this file centers on a median of 85.0, edges up from 80.0 to 87.0 over the window, and concentrates heavily: five fields hold 63% of aggregate n phds.

Social sciences lead the upper ladder, psychology and social sciences sit well above the median, and engineering trails it — a volume map of disciplines, not a prestige contest.

Sources

Data Science Learning Community. (2019). *TidyTuesday: US PhDs Awarded*. https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2019/2019-02-19/phd_by_field.csv

Editor's note

Artometrics data report from the TidyTuesday research pipeline. Charts and aggregates are reproducible from the embedded exhibits and public source files.

[View TidyTuesday source on GitHub](#)